

REV 2026

Semester 1

Lab Course

1 Credit

30 Hours

Visual Design and Smart Publishing

Course Code 1149 — Complete Polytechnic Revision 2026 Handbook

PROGRAMME	Commercial Practice
SUBJECT CODE	1149
SEMESTER	1
COURSE TYPE	Lab
TEACHING SCHEME (L:T:P:R)	0:0:2:0
CONTACT HOURS	30
CREDITS	1
CIE MARKS	40
ESE MARKS	60

Official Source Declaration

 **Source:** Kerala Polytechnic Revision 2026 Syllabus — Course 1149 (Visual Design and Smart Publishing). This handbook is prepared strictly from the official syllabus document provided. All module headings, topics, subtopics, practical outcomes, assessment schemes, and learning resources are derived directly from the source.

 **Verified Inconsistencies:** The syllabus states CIE = 40 and ESE = 60 in the course description. However, the "Assessment Methodology-Lab" table shows a "Converted to" row with values 5 (Attendance), 30 (CE), 15 (CA1), 15 (CA2), 10 (OEE), and 40 (ESE) — which sums to 75 for CIE (5+30+15+15+10), not 40. This handbook uses the stated CIE=40, ESE=60 as the official marks, and presents the assessment components as listed in the detailed tables. The "Scheme of Evaluation-Practical (Continuous Internal Evaluation)" totals 50 marks for internal assessment. The discrepancy is noted for transparency. All other content is faithfully extracted.

Course Dashboard

30

CONTACT HOURS

1

CREDIT

4

MODULES

40+60

CIE + ESE

Course Introduction

Visual design and smart publishing work together to present information clearly and attractively. Visual design uses colours, images, fonts, and layouts to make content easy to understand and pleasant to view. Smart publishing uses digital tools to create and share this content through websites, e-books, and multimedia. Together, they help communicate better, reach more people, update content quickly, and allow easy access on different devices.

Pre-requisites

- Basic Computer Skills (Secondary Education)

Teaching & Assessment Scheme

L	T	P	S	Self-learning (SS)/Week	Total Hrs/Sem	Credits C	CIE	ESE	Total
0	0	2	0	1	30	1	40	60	100

L: Lecture (1 credit), T: Tutorial (1 credit), P: Practical (0.5 credit), S: Project (0.5 credit), SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

How to Use This Handbook

1

Understand

Read each module's concepts, tool functions, and design principles.

2

Visualise

Study the workflow diagrams and see how tools are used in practice.

3

Apply

Practice with the module-wise exercises and self-learning activities.

4

Revise

Use the quick-revision cards, key concepts, and model paper before exams.

 **Tip:** Use the Print / Save as PDF button to save a printable PDF of the entire handbook. The Search box at the bottom helps you quickly find topics across all modules.

Course Outcomes & CO-PO Mapping

Course Objectives

1. To understand the basic principles and develop skills in visual design.
2. To create visually appealing and user-friendly content and use the digital tools for smart publishing.
3. To use online, e-book, and multimedia publishing platforms.
4. To enable effective creation, management, and distribution of digital content.

Course Outcomes (CO)

Course Outcomes and Bloom's Taxonomy Level

CO	Course Outcome	Bloom's Level
CO1	Construct professional page layouts for digital media using InDesign.	Apply

CO	Course Outcome	Bloom's Level
CO2	Apply Adobe Photoshop image editing tools and techniques to produce professional-quality visual results.	Apply
CO3	Develop vector-based illustrations and designs using CorelDRAW tools.	Apply
CO4	Utilize design platforms like Canva to transform ideas into visual content.	Apply

CO-PO Mapping

Course Articulation Matrix

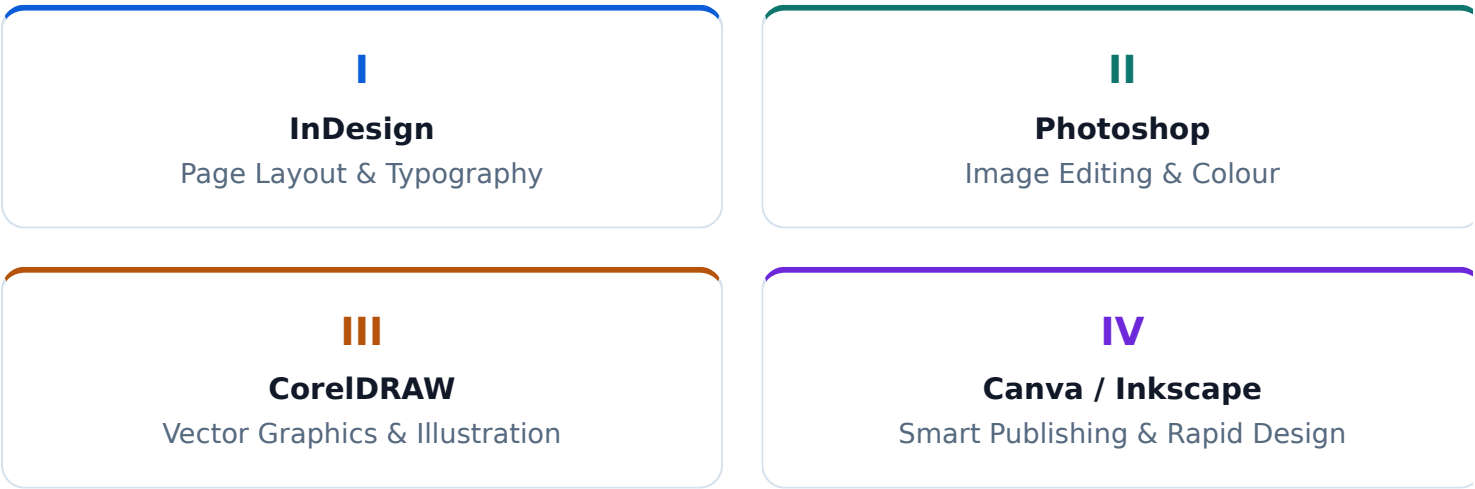
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	-	-	-	-	3	-
CO2	3	-	1	3	-	-	-	-	-	3	-
CO3	3	-	3	3	-	-	-	-	-	3	-
CO4	3	-	1	3	-	-	-	2	-	3	-
Legends: 1 – Low; 2 – Medium; 3 – High; No Correlation – “-”											

Curriculum Coverage Map

Module-wise Topics, Hours, CO & Handbook Anchor

Module	Topics	Hours	CO	Anchor
I	InDesign: Interface, Typography, Page Management, Images, Tables	6	CO1	Module I
II	Photoshop: Workspace, Tools, Colour & Tone, Image Editing	8	CO2	Module II
III	CorelDRAW: Workspace, Tools, Text Effects, Image Layout & Print	7	CO3	Module III
IV	ISM Publisher, Canva, Inkscape — Basic Design & Vector Tools	7	CO4	Module IV

How the Modules Connect



Modules I-III build foundational skills in three major design tools (InDesign, Photoshop, CorelDRAW). Module IV focuses on accessible platforms (Canva, Inkscape) for rapid content creation and smart publishing.

Tool: Adobe InDesign

- **Text frames:** Rectangular containers that hold text. Created using the Type Tool.

- **Threading:** Linking text frames so that text flows from one frame to the next. Useful for multi-column layouts and magazines.

Character & Paragraph Formatting

- **Character formatting:** Font, size, colour, kerning, tracking, baseline shift.
- **Paragraph formatting:** Alignment, indentation, spacing before/after, drop caps.
- **Font management:** Use Adobe Fonts or system fonts; check for licensing and embedding.

Text Wrap

Text wrap allows text to flow around images or shapes. Options include wrap around bounding box, wrap around object shape, and wrap offset.

Styles

- **Paragraph Styles:** Apply consistent formatting to entire paragraphs.
- **Character Styles:** Apply consistent formatting to selected characters or words.
- **Object Styles:** Apply consistent formatting to frames (fill, stroke, text wrap, etc.).

 **Tip:** Use styles to maintain consistency across long documents. Update a style once, and all instances update automatically.

Malayalam support: മലയാളം — മലയാളം മലയാളം മലയാളം മലയാളം. Text frames-
മലയാളം മലയാളം മലയാളം. Character, Paragraph Styles മലയാളം മലയാളം മലയാളം
മലയാളം മലയാളം.

1.3 Page Management — Master Pages, Grids & Alignment▼

Master Pages

Master pages are templates that define the recurring layout of your document (e.g., page numbers, headers, footers, logo). Changes made to a master page apply to all pages that use it.

- **Add, Delete, Duplicate:** Manage pages from the Pages panel.
- **Page numbering:** Insert automatic page numbers using the Type > Insert Special Character > Markers > Current Page Number.
- **Orientation:** Portrait or Landscape.

Grids, Guides & Columns

- **Guides:** Non-printing lines that help align objects. Drag from rulers (View > Show Rulers).
- **Grids:** Baseline grid for consistent text alignment; document grid for overall layout.
- **Columns:** Divide the page into vertical sections for multi-column text layouts.

Alignment & Distribution

Use the Align panel to align and distribute objects precisely. Options include align left, right, top, bottom, centre, and distribute spacing evenly.

 **Tip:** Use master pages for headers, footers, and page numbers. This saves time and ensures consistency across your document.

Malayalam support: മലയാളം അക്ഷരങ്ങൾ — അക്ഷരങ്ങൾക്കിടയിലുള്ള അകലം കൃത്യമായി നിയന്ത്രിക്കാൻ സാധിക്കും. അക്ഷരങ്ങൾ, വാക്കുകൾ, പാർഗ്രാഫുകൾ തുടങ്ങിയവയ്ക്ക് അകലം നിയന്ത്രിക്കാൻ സാധിക്കും.

1.4 Working with Images & Graphics▼

Placing Images

Use **File > Place** (or Ctrl+D) to insert images into your document. Supported formats include JPEG, PNG, TIFF, PSD, and AI.

Image Formats

- **JPEG:** Compressed, suitable for photos (lossy).
- **PNG:** Supports transparency, good for web graphics.
- **TIFF:** High-quality, suitable for print (lossless).
- **PSD:** Photoshop file, retains layers.

Scale & Fit Images

- Use the Selection Tool to resize images within frames.
- Fit content to frame, or frame to content using the Object > Fitting menu.
- Use the Links panel to manage, update, or replace linked images.

Clipping Paths

A clipping path removes the background of an image, isolating the subject. InDesign can detect and apply clipping paths from Photoshop files (PSD) or create simple paths.

Image Resolution & Quality

- **Resolution:** Measured in PPI (pixels per inch) or DPI (dots per inch). For print, use 300 PPI; for web, 72 PPI is sufficient.
- Check effective PPI in the Links panel to ensure image quality.

 **Important:** Always use high-resolution images (300 PPI) for print. Low-resolution images will appear pixelated or blurry when printed.

Malayalam support: Place ഫോട്ടോ ഫയലുകൾക്ക് ഫോട്ടോ ഫയലുകൾ. JPEG (ഫോട്ടോ ഫയലുകൾ), PNG (ഫോട്ടോ ഫയലുകൾ), TIFF (ഫോട്ടോ ഫയലുകൾ) ഫോട്ടോ ഫയലുകൾ ഫോട്ടോ ഫയലുകൾ. 300 PPI ഫോട്ടോ ഫയലുകൾ ഫോട്ടോ ഫയലുകൾ ഫോട്ടോ ഫയലുകൾ.

1.5 Tables & Advanced Text Features — PDF Export▼

Creating Tables

Use **Table > Insert Table** or draw a table using the Table tool. Tables are useful for presenting data in a structured format.

Table Styles & Formatting

- **Table Styles:** Apply consistent formatting (fill, stroke, text formatting) to tables.
- Format individual cells, rows, or columns using the Table panel.
- Merge and split cells as needed.

Advanced Text Features

- **Bullets & Numbering:** Create lists using the Paragraph panel.
- **Drop Caps:** Enlarge the first character of a paragraph.
- **Footnotes:** Add footnotes from the Type menu.

Export to PDF

Use **File > Export** to create a PDF. Choose from presets:

- **Print:** High quality for print with crop marks and bleed.
- **Digital:** Smaller file size for screen viewing, interactive features.

WORKED EXAMPLE — PDF EXPORT SETTINGS

For print PDF: Press Quality preset, include crop marks and bleed (3 mm), output CMYK colour.

For digital PDF: Smallest File Size preset, RGB colour, include hyperlinks if needed.

 **Tip:** Always check your PDF before sending to print. Use the Preflight panel to detect and fix errors.

Malayalam support: ഫോട്ടോ ഫയലുകൾ (Tables) ഫോട്ടോ ഫയലുകൾ ഫോട്ടോ ഫയലുകൾ. PDF-ഫോട്ടോ ഫയലുകൾ ഫോട്ടോ ഫയലുകൾ ഫോട്ടോ ഫയലുകൾ Press Quality, ഫോട്ടോ ഫയലുകൾ Smallest File Size ഫോട്ടോ ഫോട്ടോ ഫയലുകൾ ഫോട്ടോ ഫയലുകൾ.

Malayalam support: Photoshop — മലയാളം അക്ഷരങ്ങൾക്ക് പൂർണ്ണമായും പിന്തുണയ്ക്കുന്നു. Layers-മലയാളത്തിൽ പേര്-അക്ഷരങ്ങൾക്ക് പൂർണ്ണമായും പിന്തുണയ്ക്കുന്നു. PSD മലയാളത്തിൽ അക്ഷരങ്ങൾക്ക് പൂർണ്ണമായും പിന്തുണയ്ക്കുന്നു.

2.2 Working with Tools Panel — Selection, Retouching & Correction▼

Selection Tools

- **Marquee Tools:** Rectangular, Elliptical, Single Row/Column selections.
- **Lasso Tools:** Freehand, Polygonal, Magnetic Lasso for irregular selections.
- **Quick Selection / Magic Wand:** Select based on colour and tone.
- **Select & Mask:** Refine edges of selections, especially for hair and complex edges.

Image Retouching & Correction

- **Healing Brush & Spot Healing:** Remove blemishes, scratches, and imperfections.
- **Clone Stamp:** Copy pixels from one area to another.
- **Patch Tool:** Repair larger areas by matching texture and tone.

Painting & Drawing Tools

- **Brush Tool:** Paint with various brush shapes and sizes.
- **Pencil Tool:** For hard-edged drawing.
- **Shape Tools:** Draw rectangles, circles, lines, and custom shapes.

Filters & Effects

- **Blur:** Gaussian Blur, Motion Blur, Lens Blur.
- **Sharpen:** Sharpen, Unsharp Mask.
- **Artistic:** Watercolour, Oil Paint, etc.
- **Distort:** Liquify, Pinch, Spherize.

 **Important:** Use non-destructive editing techniques: work on duplicate layers, use adjustment layers, and use Smart Objects for filters.

Malayalam support: മറക്കുവേല (Marquee), ലാസോ (Lasso), മേജിക് വാൻഡ് (Magic Wand) തിരഞ്ഞെടുക്കൽ ഉപകരണങ്ങൾ ഉപയോഗിക്കുക. ഹീലിംഗ് ബ്രഷ്, ക്ലോൺ സ്റ്റാമ്പ് തുടങ്ങിയ റിട്ടച്ച് ഉപകരണങ്ങൾ ഉപയോഗിക്കുക.

2.3 Colour & Tone Adjustments — Modes, Brightness, Curves▼

Colour Modes

- **RGB:** Red, Green, Blue — used for digital screens (web, monitors).
- **CMYK:** Cyan, Magenta, Yellow, Black — used for print.
- **Grayscale:** Black and white images.
- Change mode via Image > Mode.

Foreground & Background Colours

The foreground colour is used for painting and filling; the background colour is used for gradient fills and the background of the canvas. Click the colour swatches to change them.

Brightness / Contrast

Adjust the overall lightness and contrast of an image. Use Image > Adjustments > Brightness/Contrast.

Levels & Curves

- **Levels:** Adjust shadows, midtones, and highlights using a histogram.
- **Curves:** More precise control over tonal range by adjusting the curve graph.

Hue / Saturation

Adjust the colour balance, saturation (vibrancy), and lightness. Use Image > Adjustments > Hue/Saturation.

WORKED EXAMPLE — COLOUR CORRECTION

Given: An underexposed image with dull colours.

Steps: 1. Add a Levels adjustment layer — move the shadow slider right and highlight slider left. 2. Add a Hue/Saturation adjustment layer — increase saturation by +20. 3. Add a Curves adjustment layer for fine-tuning contrast.

 **Tip:** Use adjustment layers (Layers > New Adjustment Layer) instead of direct adjustments. This allows you to edit or remove the adjustment later without affecting the original image.

Malayalam support: RGB (റേഡ് ഗ്രീൻ ബ്ലൂ), CMYK (സിയറിയം മേഗന്റാ ക്വട്ടർ) നിങ്ങളുടെ ചിത്രങ്ങൾക്ക് കൃത്യമായ നിറങ്ങൾ നൽകുന്നു. Levels, Curves എന്നിവ ഉപയോഗിച്ച് നിങ്ങളുടെ ചിത്രങ്ങൾക്ക് കൃത്യമായ ടോണൽ റേഞ്ച് നൽകുന്നു.

2.4 Image Creation & Basic Editing — Canvas, Crop, History▼

Opening & Saving Image Files

- **Open:** File > Open (Ctrl+O).
- **Save:** File > Save (Ctrl+S) — saves as PSD.
- **Save As:** File > Save As (Ctrl+Shift+S) — choose format (JPEG, PNG, TIFF, etc.).
- **Export:** File > Export > Save for Web — optimise for web use.

Canvas Size vs. Image Size

- **Image Size:** Changes the dimensions and resolution of the image content (scales the image).
- **Canvas Size:** Changes the work area without scaling the image. You can add or remove space around the image.

Crop & Straighten

- **Crop Tool:** Remove unwanted areas of the image. Also allows straightening.
- **Straighten:** Use the Crop Tool with the Straighten option to align images.

Undo, Redo & History

- **Undo:** Ctrl+Z (step back).
- **Redo:** Ctrl+Shift+Z (step forward).
- **History Panel:** Shows a list of recent actions. Click any state to revert to that point.

Print Setup

Use File > Print to set up printing options: paper size, orientation, scale, and colour management.

 **Tip:** The History Panel is your best friend for non-destructive experimentation. Increase the history states (Edit > Preferences > Performance) to keep more steps.

Malayalam support: Image Size vs Canvas Size — Image Size                 

- **Toolbox:** Shape, Pick, Bezier, Pen, Text, and other drawing tools.
- **Property Bar:** Context-sensitive options for the selected tool or object.
- **Docker windows:** Object Manager, Color Palette, Object Properties, etc.

Page Setup

- **Page Size:** Choose from presets (A4, A5, Letter) or custom size.
- **Orientation:** Portrait or Landscape.
- **Rulers & Grid:** Enable rulers (View > Rulers) and grid (View > Grid) for precise alignment.

Zoom, Pan & View Modes

- **Zoom:** Zoom in/out using the Zoom tool or mouse scroll.
- **Pan:** Use the Pan tool or hold the middle mouse button to drag.
- **View Modes:** Wireframe, Draft, Normal, Enhanced — choose based on performance and preview needs.

Document Management

- **Create New:** File > New (Ctrl+N).
- **Save:** File > Save (Ctrl+S) — saves as CDR (CorelDRAW native format).
- **Export:** File > Export — choose formats like PNG, JPEG, PDF, SVG.

 **Tip:** Use the Enhanced view for detailed design work. Switch to Wireframe view for faster performance on complex drawings.

Malayalam support: CorelDRAW — കേരളത്തിലെ മലയാളത്തിന് പരമാവധി പിന്തുണ നൽകുന്ന വെക്ടർ ഗ്രാഫിക്സ് പ്രോഗ്രാം. Page Size, Orientation, Rulers, Grid എന്നിവയ്ക്ക് മലയാളത്തിൽ ലഭ്യമാണ്. CDR ഫോർമാറ്റിൽ മലയാളം സാക്ഷ്യം വഹിക്കുന്നു.

3.2 Working with Tools Panel — Shapes, Node Editing, Modify▼

Shape Tools

- **Rectangle Tool:** Draw rectangles and squares.
- **Ellipse Tool:** Draw ellipses and circles.
- **Polygon Tool:** Draw regular polygons (stars, polygons).
- **Freehand Tool:** Draw freeform lines.

Bezier & Pen Tools

- **Bezier Tool:** Create precise curves and paths by placing anchor points and adjusting handles.
- **Pen Tool:** Similar to Bezier, with more options for drawing and editing paths.

Node Editing

Use the **Shape Tool** (F10) to edit nodes of curves and shapes. You can add, delete, move, and convert nodes (cusp, smooth, symmetrical).

Pick Tool & Object Modification

- **Pick Tool:** Select, move, rotate, scale, and skew objects.
- **Modify Objects:** Use the Transformation Docker for precise movement, rotation, scale, and mirror.
- **Align & Distribute:** Align objects relative to each other or to the page.

Colour, Fill & Outline

- **Fill:** Apply uniform fill, fountain fill (gradient), or pattern fill.
- **Outline:** Set outline colour, width, and style (solid, dashed, etc.).
- Use the Colour Palette (right-click for outline, left-click for fill).

 **Tip:** Node editing is key to creating custom shapes. Practice converting nodes and adjusting Bezier handles to create smooth curves.

Malayalam support: Shape Tools (Rectangle, Ellipse, Polygon) ഫോറം ഫോറം ഫോറം. Bezier, Pen ഫോറം ഫോറം ഫോറം ഫോറം. Node Editing ഫോറം ഫോറം ഫോറം.

3.3 Working with Text Effects — Artistic Text, Path, Effects▼

Artistic Text vs. Paragraph Text

- **Artistic Text:** Single line or short phrases, can be scaled and styled like an object. Used for logos, titles, and headings.
- **Paragraph Text:** Multi-line text in a frame. Used for body text, articles, and long passages.

Text Formatting Tools

- **Font, size, style:** Select from the Property Bar.
- **Character formatting:** Kerning, tracking, baseline shift.
- **Paragraph formatting:** Alignment, indentation, spacing, bullet lists.

Text on Path

Use the **Text on Path** tool (or Text > Fit Text to Path) to make text follow a curve or shape. This is useful for circular logos and creative typography.

Text to Curves

Convert text to curves (Ctrl+Q) to edit individual characters as vector shapes. This ensures the font remains unchanged even if the recipient doesn't have the font installed.

Effects Tools

- **Shadow:** Apply drop shadows to text or objects.
- **Transparency:** Make objects partially transparent (uniform, fountain, or pattern transparency).
- **Contour:** Create concentric outlines of a shape or text.
- **Blend:** Create a blend between two objects.

! Important: When sending a CorelDRAW file for print, convert all text to curves to avoid font substitution issues. Keep a copy of the original file with editable text for future modifications.

Malayalam support: Artistic Text (രീതി, അക്ഷരങ്ങൾ), Paragraph Text (പാർഗ്രാഫ് അക്ഷരങ്ങൾ). Text on Path — രേഖാമൂലം അക്ഷരങ്ങൾ അടങ്ങിയവ. Effects (Shadow, Transparency, Contour) അക്ഷരങ്ങൾക്ക് അനുയോജ്യമായവ.

3.4 Image Layout & Print — Bitmap Editing, Shaping, Print Settings▼

Bitmap Images

CorelDRAW can import and edit bitmap images (JPEG, PNG, TIFF, etc.) using **File > Import**.

Bitmap Editing

- Use the **Edit Bitmap** option to open the image in Corel PHOTO-PAINT for advanced editing.
- Basic bitmap tools: Crop, Resample, and Colour Adjustments.

Shaping Tools (Weld, Trim, Intersect)

- **Weld:** Combine two or more shapes into a single shape.
- **Trim:** Cut one shape using another shape.
- **Intersect:** Create a shape from the overlapping area of two shapes.
- These are found in the Property Bar when multiple objects are selected.

Print Settings

- **Bleed, Margins & Crop Marks:** Set up print-ready files with appropriate bleed (3–5 mm), margins, and crop marks.
- **Page Setup > Print:** Choose printer, paper size, and orientation.
- **Print Preview:** Check the layout before printing.

Export Formats

- **PDF:** Best for print and sharing (preserves vector quality).
- **PNG:** For web and transparency.
- **JPEG:** For photos and web.

- **SVG:** For web and scalable graphics.

WORKED EXAMPLE — CREATING A LOGO

Steps: 1. Draw a circle using the Ellipse tool. 2. Add text using Artistic Text. 3. Fit text to path using Text on Path. 4. Add a star shape using Polygon tool. 5. Weld shapes together. 6. Apply fill and outline colours. 7. Export as PNG and PDF.

Malayalam support: Weld, Trim, Intersect രണ്ടു വക്രങ്ങളെ ഒരുമിച്ച് വക്രമാക്കുക രണ്ടു വക്രങ്ങളെ തമ്മിൽ മുറിക്കുക രണ്ടു വക്രങ്ങളുടെ ഓരോ ഭാഗത്തെയും വക്രമാക്കുക
രണ്ടു വക്രങ്ങളുടെ ഓരോ ഭാഗത്തെയും വക്രമാക്കുക രണ്ടു വക്രങ്ങളുടെ ഓരോ ഭാഗത്തെയും വക്രമാക്കുക. രണ്ടു വക്രങ്ങളുടെ ഓരോ ഭാഗത്തെയും വക്രമാക്കുക Bleed, Crop Marks രണ്ടു വക്രങ്ങളുടെ ഓരോ ഭാഗത്തെയും വക്രമാക്കുക.

 **Module III Summary:** CorelDRAW is for vector graphics. Key features: Shape tools, Bezier/ Pen for custom paths, Node editing for precision, Artistic and Paragraph text, Text on Path, Effects (Shadow, Transparency), Shaping tools (Weld, Trim, Intersect), and print-ready export.

MODULE IV

ISM Publisher, Canva & Inkscape — Smart Publishing

Learn to use accessible design platforms: ISM Publisher for Malayalam typing, Canva for rapid design, and Inkscape for vector thumbnails.

Hours: 7

CO: CO4

Bloom: Apply

Tools: ISM, Canva, Inkscape

4.1 Getting Started with ISM Publisher — Malayalam Typing▼

ISM Publisher (or similar Malayalam typing software) enables typing and publishing in Malayalam using the Inscript keyboard layout.

Basic Settings

- **Select Script:** Choose Malayalam (Inscript) from the language settings.
- **Tune Application:** Configure the software for optimal performance — set font preferences and keyboard shortcuts.
- **Keyboard Modes:** Switch between Inscript (standard Malayalam keyboard) and English keyboard modes as needed.
- **Font Settings:** Choose a suitable Malayalam font (e.g., Karthika, Rachana, or Noto Sans Malayalam).

Typing in Malayalam

- **Inscript layout:** The standard keyboard layout for typing in Indian languages. Each key corresponds to a Malayalam character.

- **English mode:** Type in English within the same document when needed (e.g., for technical terms or URLs).

Applications

ISM Publisher is useful for creating Malayalam documents, posters, brochures, and digital content for regional audiences.

 **Tip:** Practice typing common Malayalam words and phrases to become familiar with the Inscript layout. Use the on-screen keyboard for reference.

Malayalam support: ISM Publisher — മലയാളം അക്ഷരങ്ങൾക്ക് അനുയോജിക്കുന്ന അക്ഷരസമൂഹം. Inscript അക്ഷരസമൂഹം ഉപയോഗിച്ച് മലയാളം അക്ഷരങ്ങൾ എഴുതാൻ സാധിക്കും. English, Malayalam അക്ഷരങ്ങൾ ഉപയോഗിച്ച് എഴുതാൻ.

4.2 Canva — Basic Design Creation▼

Canva is a web-based graphic design platform that makes design accessible to non-designers. It offers templates, drag-and-drop tools, and a vast library of elements.

Interface & Basic Tools

- **Home:** Browse templates by category (social media, posters, presentations, etc.).
- **Design Editor:** Drag-and-drop interface with tools for text, elements, and backgrounds.
- **Search & Filter:** Find templates, photos, icons, and illustrations using the search bar. Filter by colour, style, or type.

Page Size & Layout

- Choose from preset sizes (Instagram post, A4, Facebook cover, etc.) or create a custom size.
- Use **Layouts** to quickly arrange elements in different configurations.

Typography, Colours & Alignment

- **Fonts:** Hundreds of free fonts available. Use font combinations for headers and body text.
- **Colours:** Use the colour picker to match brand colours or create a cohesive palette.
- **Alignment:** Use the alignment tools to position elements precisely.

Elements

- **Shapes:** Rectangles, circles, lines, and more.
- **Icons:** Thousands of icons for visual communication.
- **Photos:** Access stock photos from the Canva library or upload your own.

WORKED EXAMPLE — DESIGN A QUOTE POSTER

Steps: 1. Search for "quote poster" template. 2. Add a background colour or image. 3. Add your quote with a stylised font. 4. Add a shape or icon for emphasis. 5. Adjust alignment and spacing. 6. Download as PNG or PDF.

 Tip: Canva's templates are a great starting point. Customise them with your own colours, fonts, and content to create unique designs.

Malayalam support: Canva — ഐക്യം ഐക്യത്തിനായി എന്നുള്ളതാണ് ഐക്യം എന്നുള്ളതാണ്. Templates, drag-and-drop ഐക്യം ഐക്യത്തിനായി എന്നുള്ളതാണ് ഐക്യം എന്നുള്ളതാണ്. Fonts, Colours, Elements ഐക്യം ഐക്യത്തിനായി എന്നുള്ളതാണ്.

4.3 Working with Vector Tools — Canva & Inkscape (YouTube Thumbnail)▼

This topic combines Canva and Inkscape for creating vector-based designs, specifically a YouTube thumbnail.

YouTube Thumbnail Design (Canva)

- **Template:** Search for "YouTube Thumbnail" in Canva's template library.
- **Background:** Add a background colour or upload an image.
- **Icons & Pictures:** Add small icons or pictures for emphasis and visual interest.
- **Text:** Use large, bold, and readable text to convey the video title or key message.

Creating a Thumbnail in Inkscape

Inkscape is a free and open-source vector graphics editor, similar to CorelDRAW and Adobe Illustrator.

- **Page Setup:** Set page size to 1280 × 720 pixels (standard YouTube thumbnail size).
- **Text:** Add large, readable text. Use bold fonts and contrasting colours.
- **Images & Shapes:** Import images or draw simple shapes to enhance the design.
- **Save:** Export as PNG (File > Export PNG Image) for use on YouTube.

WORKED EXAMPLE – YOUTUBE THUMBNAIL IN INKSCAPE

Steps: 1. File > Document Properties > Page size: 1280 × 720 px. 2. Add a rectangle for the background (fill with colour or gradient). 3. Use the Text tool to add a title with a bold font. 4. Import an image or draw a simple shape using the Shape tools. 5. Align elements and adjust colours. 6. File > Export PNG Image to save.

 **Important:** YouTube thumbnails need to be eye-catching and readable at small sizes. Use high-contrast colours, large text, and clear visuals.

Malayalam support: YouTube Thumbnail — 1280 × 720 പരമാവധി വീഡിയോ ഫ്രെയിം. Inkscape പരമാവധി വീഡിയോ ഫ്രെയിം. പരമാവധി, പരമാവധി വീഡിയോ ഫ്രെയിം.

 **Module IV Summary:** ISM Publisher enables Malayalam typing and publishing. Canva offers rapid design with templates and a vast library of elements. Inkscape is a free vector tool for creating graphics like YouTube thumbnails. Together, these tools support smart publishing for diverse audiences.

Key Design Concepts

Typography — Font Selection

Serif vs. Sans Serif

Serif: traditional, formal (e.g., Times). Sans-serif: modern, clean (e.g., Arial).

Colour — RGB vs CMYK

RGB = Screen | CMYK = Print

RGB: Red/Green/Blue — additive. CMYK: Cyan/Magenta/Yellow/Black — subtractive.

Image Resolution

Print: 300 PPI | Web: 72 PPI

Higher PPI = better quality, larger file size.

Bleed (Print)

3–5 mm

Extra area beyond trim to allow for cutting.

Aspect Ratio (YouTube Thumbnail)

1280 × 720 px (16:9)

Standard YouTube thumbnail size.

File Formats

JPEG (photo) | PNG (transparency) | PDF (print)

Choose based on purpose.

Diagram Library for Rapid Revision

InDesign Workspace

Module I — Tools Panel, Control Panel, Panels (Pages, Layers, Swatches).

Photoshop Layers

Module II — Layer management, styles, opacity, blending modes.

CorelDRAW Tools

Module III — Shape, Bezier, Pick, Text, Node Editing.

Canva Interface

Module IV — Templates, Elements, Text, Colours.

Inkscape Thumbnail

Module IV — 1280×720 setup, text, shapes, PNG export.

ISM Publisher

Module IV — Malayalam typing, Inscript layout.

Self-Learning Activities (Official)

Module I — InDesign Basics

1 hour

Identify Selection Tool, Text Tool, Rectangle & Shape Tools. Create a new document and share it.

Module I — Quote Poster

1 hour

Create a mini "quote poster" with your favourite quote and display on the notice board.

Module I — Abstract Poster

1 hour

Make a simple "abstract poster" using only shapes and colours and share it.

Module I — Book Cover Redesign

1 hour

Redesign a "Book Cover" from your favourite book and share it.

Module II — Quote Image

1 hour

Create a "quote image" with a favourite saying and share.

Module II — Photo Brightness

1 hour

Edit a photo to make it brighter and more colourful and share in Google Drive.

Module II — Background Removal

1 hour

Remove the background of an image and replace it.

Module III — Outline/Stroke

1 hour

Apply outline/stroke to shapes and share it.

Module III — Quote Card/Nameplate

1 hour

Design a "quote card" or 'nameplate' and share it.

Module III — Alignment Exercise

1 hour

Use alignment tools to arrange objects neatly and make a copy.

Module III — School Club Logo

1 hour

Design a "school club logo" or "event poster" and display on the department notice board.

Module IV — Malayalam Brochure

1 hour

Design a "simple brochure" or 'flyer' using Malayalam font and share it.

Module IV — Canva Templates

1 hour

Use Canva templates to create Posters or digital flashcards or Make a timeline.

Lab Exam Preparation

1 hour (x2)

Prepare for lab exams (Module II, Module IV).

Total Self-Learning Hours: 15

Assessment Methodology

Assessment Breakdown — Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE	Total
Continuous evaluation	5	30	15	15	10	—	—
Practical test (first half)							
Practical test (second half)	—	—	—	—	—	—	—
Open-ended experiment	—	—	—	—	—	—	—
Practical (Duration 3h)	—	—	—	—	—	40	40
Converted to	5	30	15	15	10	40	—
Total marks	—	—	—	—	—	—	100
CIE: 40 marks (Attendance 5 + CE 30 + CA1 15 + CA2 15 + OEE 10) ESE: 60 marks Total: 100							

Scheme of Evaluation — Practical (CIE)

- **Preparation & Punctuality:** 10 marks
 - **Tool Usage & Practical Procedure:** 10 marks
 - **Design Execution & File Management:** 5 marks
 - **Output Quality & Design Interpretation:** 10 marks
 - **Viva Voce / Concept Understanding:** 10 marks
 - **Workmanship, Discipline & Creativity:** 5 marks
- Total: 50 marks**

Scheme of Evaluation — ESE (Practical)

- **Write-up / Procedure:** 10 marks
 - **Practical Execution & Tool Usage:** 10 marks
 - **Output / Result:** 8 marks
 - **Viva Voce:** 8 marks
 - **Record:** 4 marks
- Total: 40 marks**

Open-Ended Experiment (OEE)

- **Originality & Relevance:** 10 marks
 - **Clarity of Objectives & Plan:** 10 marks
 - **Correctness of Execution & Recording:** 10 marks
 - **Analysis, Interpretation & Presentation:** 10 marks
 - **Teamwork & Innovation:** 10 marks
- Total: 50 marks**

Series Test & ESE Pattern

- **Series Test I:** Modules I & II (1.5 hrs)
- **Series Test II:** Modules III & IV (1.5 hrs)
- **ESE Split-up:**
 - Theory / Concept (all modules) — 10 marks
 - Practical Design Task — 18 marks
 - Viva Voce — 8 marks
 - Record — 4 marks

ESE Total: 40 marks

 **Note:** The official CIE is 40 marks, but the Assessment Methodology table shows component marks summing differently. Refer to the Source Declaration for details. Students must submit a certified practical record with outputs from all modules for ESE.

Module-wise Questions with Answers

Module I — InDesign

Q1: What are the key panels in the InDesign workspace? What is their purpose?▼

Answer: Key panels in InDesign include:

- **Pages Panel:** Manage pages, add/delete/duplicate, apply master pages, and set page numbering.
- **Layers Panel:** Organise content into layers for better control over visibility and stacking order.
- **Links Panel:** Manage linked images and other assets. Check status and update links.
- **Swatches Panel:** Manage colours, gradients, and tints for consistent colour usage.
- **Character & Paragraph Panels:** Format text with fonts, sizes, alignment, and spacing.

These panels help streamline the design workflow and maintain consistency.

Q2: What is a master page? How do you apply page numbering in InDesign?▼

Answer: A **master page** is a template that defines the recurring layout of a document. It contains elements that appear on every page that uses that master page (e.g., headers, footers, logos, page numbers).

To apply page numbering:

1. Double-click the master page in the Pages panel.
2. Create a text frame where you want the page number.
3. Choose Type > Insert Special Character > Markers > Current Page Number. A placeholder (e.g., "A" for the master page prefix) appears.

4. Format the text as desired.
5. When applied to document pages, the placeholder shows the actual page number.

Master pages save time and ensure design consistency across multi-page documents.

Q3: Explain the difference between JPEG, PNG, and TIFF image formats. When would you use each?▼

Answer:

- **JPEG (Joint Photographic Experts Group):** Compressed format, suitable for photographs and web use. Lossy compression reduces file size but can affect quality. Best for web, email, and social media.
- **PNG (Portable Network Graphics):** Supports transparency (alpha channel) and lossless compression. Good for web graphics, logos, and images with transparent backgrounds.
- **TIFF (Tagged Image File Format):** High-quality, lossless format. Supports layers and high colour depth. Used for print, professional photography, and archiving.

Use cases: JPEG for photos on the web; PNG for graphics with transparency; TIFF for print and high-quality archiving.

Module II — Photoshop

Q4: What are adjustment layers in Photoshop? Why are they useful?▼

Answer: Adjustment layers are non-destructive layers that apply colour and tonal adjustments to the layers below them. Examples include Levels, Curves, Hue/Saturation, Brightness/Contrast, and Color Balance.

Benefits:

- **Non-destructive:** The original image remains unchanged. You can hide, delete, or edit the adjustment layer at any time.
- **Reusable:** Adjustment layers can be copied, moved, or applied to multiple layers using clipping masks.
- **Maskable:** You can use layer masks to apply the adjustment to only part of the image.
- **Reversible:** You can change the settings of the adjustment layer at any time without starting over.

This makes adjustment layers essential for professional, flexible editing workflows.

Q5: What is the difference between RGB and CMYK colour modes? When should you use each?▼

Answer:

- **RGB (Red, Green, Blue):** Additive colour model used for digital screens (monitors, phones, projectors). Creates colours by combining red, green, and blue light. Larger colour gamut (more vibrant colours).
- **CMYK (Cyan, Magenta, Yellow, Black):** Subtractive colour model used for print. Creates colours by subtracting light from white paper using cyan, magenta, yellow, and black inks. Smaller colour gamut.

When to use: Use **RGB** for web, social media, and any digital display. Use **CMYK** for print materials (brochures, posters, magazines). Always convert to CMYK before sending to a printer.

Q6: Explain the Levels and Curves adjustments. Which one gives more control?▼

Answer:

- **Levels:** Adjusts the tonal range of an image using a histogram. It has three sliders: black point (shadows), midtones (gamma), and white point (highlights). It's useful for quick contrast and brightness fixes.
- **Curves:** Provides more precise control over the tonal range by adjusting a curve graph. You can add multiple control points to fine-tune shadows, midtones, and highlights independently. Curves also allow per-channel adjustments (e.g., adjusting only the red channel).

Conclusion: **Curves** gives more control and is preferred for professional colour correction. Levels is simpler and good for quick adjustments.

Module III — CorelDRAW

Q7: What is the difference between Artistic Text and Paragraph Text in CorelDRAW?▼

Answer:

- **Artistic Text:** Created with the Text tool (F8) by clicking on the page. It is treated as an object — you can scale, rotate, skew, and apply effects to it. Used for single lines, titles, logos, and short phrases. Artistic Text can be converted to curves for further editing.
- **Paragraph Text:** Created by dragging a text frame with the Text tool. It flows within a frame and supports multi-line, multi-column text. Used for body text, articles, and long passages. Supports text wrap and has more formatting options (paragraph spacing, indentation, bullets, etc.).

Use cases: Artistic Text for headlines and logos; Paragraph Text for body content.

Q8: Explain the Weld, Trim, and Intersect shaping operations in CorelDRAW.▼

Answer: These are shaping tools used to combine or modify vector objects.

- **Weld:** Combines two or more objects into a single, unified shape. The resulting shape takes the fill and outline of the target object. Useful for creating custom shapes from multiple objects.
- **Trim:** Cuts one object using another object as a "cookie cutter." The overlapping area of the source object is removed. The target object remains, and the source object is discarded.
- **Intersect:** Creates a new shape from the overlapping area of two or more objects. The original objects are discarded. Useful for creating complex shapes from intersections.

These operations are found in the Property Bar when multiple objects are selected, or through the Shaping docker (Arrange > Shaping).

Q9: Why should you convert text to curves before sending a CorelDRAW file to print? ▼

Answer: Converting text to curves (Ctrl+Q) transforms text characters into vector shapes.

Reasons:

- **Font substitution:** If the recipient doesn't have the exact font installed, the text may display incorrectly. Converting to curves ensures the text appears exactly as designed.
- **Consistency:** The design stays consistent across different systems and printers.
- **Editing:** Once converted to curves, you can edit individual characters as vector shapes (node editing), allowing for custom typography.

 **Important:** Keep a copy of the original file with editable text for future modifications. Converting to curves is irreversible (though you can undo if you haven't saved).

Module IV — Canva, Inkscape, ISM Publisher

Q10: What are the advantages of using Canva for design creation? ▼

Answer: Canva is a popular web-based design platform with several advantages:

- **Ease of use:** Drag-and-drop interface makes it accessible to non-designers.
- **Templates:** Thousands of pre-designed templates for social media, posters, presentations, and more.
- **Library:** Extensive library of free photos, icons, shapes, and fonts.
- **Collaboration:** Real-time collaboration and sharing options.
- **Multi-platform:** Works on web, desktop, and mobile (iOS/Android).
- **Export options:** Download designs as PNG, JPEG, PDF, or MP4 (video).
- **Brand tools:** Brand Kit for consistent use of colours, fonts, and logos across designs.

Canva is ideal for rapid design, social media content, and projects that don't require advanced print production features.

Q11: How do you set up a YouTube thumbnail in Inkscape? What are the key design considerations?▼

Answer: YouTube thumbnails should be designed at 1280 × 720 pixels (16:9 aspect ratio).

Steps in Inkscape:

1. File > Document Properties > Page size: 1280 × 720 px.
2. Add a background (rectangle with fill colour or gradient).
3. Add text using the Text tool — use large, bold, readable fonts.
4. Import or draw images/shapes to enhance visual appeal.
5. Align elements using the Align and Distribute panel.
6. File > Export PNG Image to save as PNG.

Key design considerations:

- **Readability:** Text should be readable at small sizes (on mobile). Use high-contrast colours.
- **Simplicity:** Keep the design clean and focused on the main message.
- **Branding:** Use consistent colours, fonts, and logos.
- **Emotion:** Use expressive faces or bold text to evoke curiosity.
- **Size:** Ensure all elements fit within the 1280×720 canvas.

Q12: What is the Inscript keyboard layout? Why is it used for Malayalam typing?▼

Answer: **Inscript** is a standard keyboard layout for typing in Indian languages, including Malayalam, Hindi, Tamil, and others. It was developed by the Government of India to provide a uniform keyboard layout across all Indian languages.

Why use Inscript for Malayalam:

- **Standardisation:** The same physical key positions are used for all Indic scripts, making it easier to switch between languages.
- **Phonetic mapping:** Keys are arranged based on the phonetic sound of the characters, making it easier to learn.
- **Widely supported:** Most Malayalam typing software (ISM Publisher, Google Input Tools) supports the Inscript layout.
- **Government standard:** Inscript is the official standard for Indian language typing.

ISM Publisher supports Inscript and allows switching between Malayalam and English modes within the same document.

Practice Model Paper

Based on the official pattern. This is a practice paper, not an official SITTTTR model question paper.

End Semester Examination (ESE) — Practice Paper

Duration: 3 hours | **Total Marks:** 40

Part A — Theory (All Modules)

Answer any **5** of the following. Each question carries 2 marks. ($5 \times 2 = 10$ marks)

1. What is a master page in InDesign? State its purpose.
2. Explain the difference between RGB and CMYK colour modes.
3. What are adjustment layers in Photoshop? Why are they useful?
4. What is the difference between Artistic Text and Paragraph Text in CorelDRAW?
5. Explain the Weld and Trim operations in CorelDRAW.
6. What is the Inscript keyboard layout? Why is it used?

Part B — Practical Design Task

Answer any **1** of the following. Each question carries 18 marks. ($1 \times 18 = 18$ marks)

1. **InDesign:** Create a one-page brochure for a college event. Include a title, event details (date, time, venue), a logo, and an image. Use appropriate typography, colours, and layout. Export as PDF.
2. **Photoshop:** Edit a given image: correct the colour using Levels/Curves, remove a blemish using the Healing Brush, add a text overlay with a quote, and save as PSD and JPEG.
3. **CorelDRAW:** Design a logo for a school club. Use shapes, text, and colours. Apply effects (shadow, transparency). Save as CDR and export as PNG and PDF.
4. **Canva/Inkscape:** Create a YouTube thumbnail (1280×720) for a "DIY" video. Use large text, a background image, and icons. Export as PNG.

Part C — Viva Voce & Record

- **Viva Voce:** 8 marks — Questions based on tools, concepts, and file formats from all modules.
- **Record:** 4 marks — Completion and neatness of practical record, certified by faculty, with outputs from all modules.

Total: $10 + 18 + 8 + 4 = 40$ marks

Quick Revision

Module I — InDesign

- Tools Panel: Selection, Text, Shape, Frame.
 - Master Pages: templates for headers, footers, page numbers.
 - Typography: Character & Paragraph Styles, Text Wrap.
 - Images: Place, Links panel, resolution (300 PPI for print).
 - Tables: Insert, format, styles.
 - Export: PDF for print (Press Quality) or digital (Smallest File Size).
-

Module II — Photoshop

- Layers: non-destructive editing, Layer Styles, Opacity, Blending Modes.
- Selection Tools: Marquee, Lasso, Magic Wand, Quick Selection.
- Retouching: Healing Brush, Clone Stamp, Patch.
- Colour Modes: RGB (screen), CMYK (print).
- Adjustments: Levels, Curves, Hue/Saturation, Brightness/Contrast.
- History Panel: undo/redo, revert to previous states.

Module III — CorelDRAW

- Shape Tools: Rectangle, Ellipse, Polygon, Freehand.
- Bezier & Pen: custom paths and curves.
- Node Editing: Shape tool (F10) — add, delete, convert nodes.
- Artistic Text (titles) vs Paragraph Text (body).
- Text on Path, Text to Curves.
- Shaping: Weld, Trim, Intersect.
- Effects: Shadow, Transparency, Contour.

Module IV — Canva / Inkscape / ISM

- ISM Publisher: Malayalam typing (Inscript layout).
- Canva: Templates, Elements, Fonts, Colours, Export.
- Inkscape: free vector tool, 1280×720 for YouTube thumbnails.
- YouTube Thumbnail: large text, high contrast, simple design.
- Smart Publishing: digital-first, accessible, multi-platform.

Common Mistakes to Avoid

- ❌ Using low-resolution images (72 PPI) for print — always use 300 PPI.
- ❌ Forgetting to set bleed (3–5 mm) for print documents.
- ❌ Not converting text to curves before sending CorelDRAW files to print.
- ❌ Editing directly on the background layer in Photoshop (no layers).
- ❌ Not using adjustment layers for non-destructive editing.
- ❌ Using RGB colour mode for print — convert to CMYK.
- ❌ Overloading designs with too many fonts or colours — keep it simple.
- ❌ Forgetting to save as PSD (Photoshop) or CDR (CorelDRAW) to retain editability.
- ❌ Using the wrong page size for YouTube thumbnails (not 1280×720).

Exam Checklist

- ✅ Know the interface and key panels for each software.
- ✅ Understand file formats: PSD, JPEG, PNG, TIFF, CDR, PDF, SVG.
- ✅ Practice creating a poster/brochure in InDesign.
- ✅ Practice colour correction and retouching in Photoshop.

-  Practice logo design and text effects in CorelDRAW.
-  Practice creating a YouTube thumbnail in Canva or Inkscape.
-  Be able to type in Malayalam using ISM Publisher (Inscript layout).
-  Understand print settings: bleed, crop marks, CMYK, PDF export.
-  Maintain a neat practical record with all exercises and outputs.

Glossary

Key Terms and Meanings

Term	Meaning
Visual Design	The use of colour, typography, and layout to communicate effectively.
Smart Publishing	Digital creation and distribution of content across multiple platforms.
Typography	The art and technique of arranging type for readability and visual appeal.
Master Page	A template page in InDesign that defines recurring layout elements.
Bleed	Extra area beyond the trim edge (3-5 mm) for full-bleed printing.
RGB	Red, Green, Blue — additive colour model for digital screens.
CMYK	Cyan, Magenta, Yellow, Black — subtractive colour model for print.
PPI (Pixels Per Inch)	Resolution measure for images on screen. 300 PPI is standard for print.
DPI (Dots Per Inch)	Resolution measure for printers.
Adjustment Layer	Non-destructive colour/tonal adjustment layer in Photoshop.
Layer Style	Effects applied to a layer (drop shadow, glow, stroke, etc.).
Artistic Text	Single-line text treated as an object in CorelDRAW.
Paragraph Text	Multi-line text in a frame in CorelDRAW.
Weld / Trim / Intersect	Shaping operations in CorelDRAW to combine or modify objects.
Inscript Layout	Standard Indian language keyboard layout (including Malayalam).
Canva	Web-based graphic design platform with templates and drag-and-drop tools.
Inkscape	Free, open-source vector graphics editor.

Learning Resources

Textbooks

Recommended Textbooks

Sl. No.	Title	Author	Publication
1	Simplified DTP Course Book	Singh Vishnu P.	Asian Publishers
2	Learning Desk Top Publishing	Ramesh Bangia	Khanna Publishers
3	Introduction to Desktop Publishing with Adobe InDesign	Indranil Roy	Notion Press

Web Resources

-
- CorelDRAW Tutorials
-

- ... and ' + (targets.length - limit) + ' more.'; searchResults.innerHTML = html; const links = searchResults.querySelectorAll('a'); links.forEach(function(a) { a.addEventListener('click', function(e) { const href = a.getAttribute('href'); if (href && href.startsWith('#')) { const target = document.querySelector(href); if (target) { e.preventDefault(); const details = target.closest('details'); if (details && !details.hasAttribute('open')) details.setAttribute('open', ''); setTimeout(function() { target.scrollIntoView({ behavior: 'smooth' }); }, 100); } } }); }); if (searchButton) searchButton.addEventListener('click', performSearch); if (searchInput) { searchInput.addEventListener('keydown', function(e) { if (e.key === 'Enter') performSearch(); }); } if (clearSearch) { clearSearch.addEventListener('click', function() { if (searchInput) { searchInput.value = ''; searchResults.innerHTML = ''; } }); } // ===== KEYBOARD SHORTCUT: Ctrl+/ = focus search ===== document.addEventListener('keydown', function(e) { if ((e.ctrlKey || e.metaKey) && e.key === '/') { e.preventDefault(); if (searchInput) searchInput.focus(); } }); // ===== RESPECT REDUCED MOTION ===== const prefersReduced = window.matchMedia('(prefers-reduced-motion: reduce)'); if (prefersReduced.matches) { document.querySelectorAll('.rotating, .rotating.slow, .rotating.fast').forEach(function(el) { el.classList.remove('rotating', 'slow', 'fast'); }); } // ===== OPEN TOPICS ON INTERNAL LINK ===== document.addEventListener('click', function(e) { const a = e.target.closest('a'); if (!a) return; const href = a.getAttribute('href'); if (href && href.startsWith('#')) { const targetId = href.substring(1); const target = document.getElementById(targetId); if (target) { const details = target.closest('details'); if (details && !details.hasAttribute('open')) { details.setAttribute('open', ''); } } } }); });